Revenue and Business Model

Customers, pricing, a three-year plan and the risks

CoolRoute: Pedestrian Routing with Shade and Heat-Exposure Budgets

> All money figures below are **planning assumptions** for discussion. They are not forecasts and have not been tested with customers. Where we say 'to validate', the next step is to ask real buyers.

1. Who needs this

Customer	Pain	What they get
City and smart-city offices	Cannot see which streets are heat traps; need to decide where to plant trees and add shelters	A heat map of walking routes and a tool to test shade plans
Citizens (older people, parents, daily walkers)	Hot, tiring walks; steps and long waits at crossings	Free cooler routes, step-free option
Delivery, ride and tourism apps	Walking legs are unpleasant in summer; lost orders and bad reviews	A route API that adds a cool-walk option
Universities, campuses, hospitals, malls	Safe and comfortable walking between buildings	Campus routing and shade audit

2. How it earns money

Stream	Who pays	Price assumption	To validate
City licence	City or ward office	Rs 6 lakh per city per year in year 1, rising to Rs 10 lakh with more features	Ask two smart-city offices what they pay for similar tools
Route API	Delivery, tourism and mobility apps	Rs 0.15 per route call	Compare with map-API pricing and test willingness to pay
Premium app	Citizens	Rs 399 per year (about Rs 33 a month) for saved places, alerts, audio guidance	Run a survey and a small pilot
Free tier	Nobody	Basic cool route for everyone	Builds trust and data; keeps the public-good mission

3. Three-year plan (illustrative)

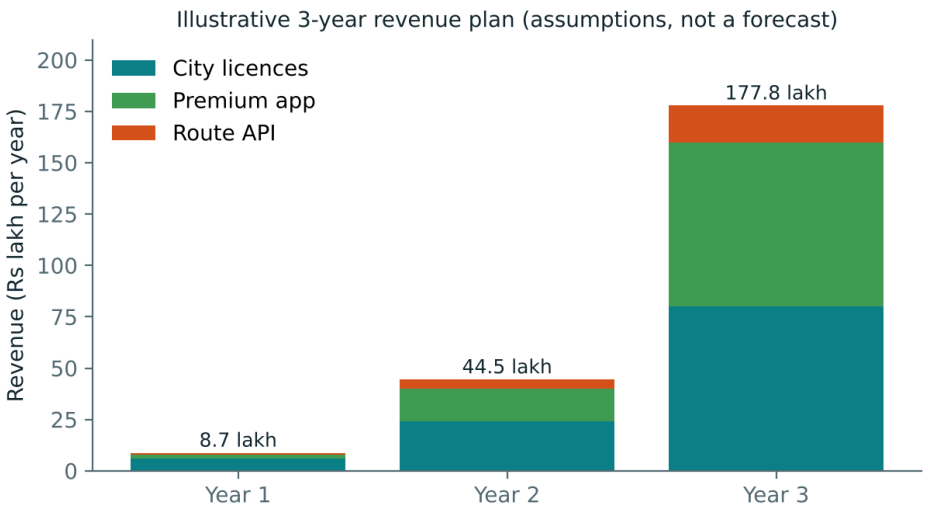


Figure 1. Revenue by stream under the assumptions in the table below.

Assumption	Year 1	Year 2	Year 3
Cities paying	1 at Rs 6 lakh	3 at Rs 8 lakh	8 at Rs 10 lakh
API calls per year	0.5 million	3 million	12 million
Premium subscribers	500	4,000	20,000
Revenue (Rs lakh)	**8.7**	**44.5**	**177.8**

4. Costs stay low

- Open data (OpenStreetMap, open weather, free satellite imagery) and open-source tools keep data costs near zero at the start.
- The engine is light. One small server can handle a city-size graph with caching.
- Main costs later: a part-time data engineer, hosting, support for city pilots, and survey and field validation.
- Cost per route call is tiny compared with the assumed price, so the API has room for a healthy margin once volume grows (to validate with real hosting bills).

5. Go-to-market

Phase	Action	Proof we want
0 to 3 months	Pilot on the Manipal University Jaipur campus and one nearby ward; walk the routes and compare with the model	Students and staff prefer the cool route; model matches what people feel
3 to 9 months	Offer a free pilot to a city office with a shade-planning report	One signed paid licence
9 to 18 months	Open the route API to two partner apps	API calls growing every month
18 months on	Premium app and more cities	Subscribers and repeat renewals

6. How we compare

	Normal map apps	CoolRoute
Optimises	Distance and time	Distance, time, shade, heat budget, effort
Time-of-day heat	No	Yes, through the sun model

	Normal map apps	CoolRoute
Step-free as a hard rule	Sometimes as a filter	Yes, always a hard rule
Says when data is weak	Rarely	Yes, with warnings
Explains the trade-off	No	Yes, one plain sentence

7. Impact we can measure

- Heat exposure avoided per trip (sun-metres), shown in the app.
- Share of walking done in shade.
- Number of step-free routes found that other tools miss.
- For cities: the list of streets that cause most heat exposure, which is where trees and shelters help most.

8. Risks and answers

Risk	Answer
Shade data is wrong in the real city	Confidence values, dashed streets and warnings; field checks in the pilot
People read the score as a medical warning	Wording and a note on every result: a comparison score, not a health risk
Free data or API terms change	Self-host OpenStreetMap extracts; keep the engine independent of any one provider
Big map companies copy the idea	Move first with city partnerships and local data; open, explainable method
Cities buy slowly	Start with campuses and the free tier; use pilots as proof